EXPERIMENT - 02

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Logical, Bit-Manipulation and Shift Operations Using 8-bit and 16-bit Data Manipulation

a) Verification of Bit Masking Operations Using 8-Bit Processor Assembly Language Programming

b) Testing of Bit Setting and Clearing Operations Using 8-Bit Assembly Language Programming

OBJECTIVE

To develop and verify assembly language programs for bit masking, bit setting, and bit clearing operations using the 8085 8-bit microprocessor.

ALGORITHM

1. Initialize the memory locations with the input data.
2. Load the input data into the accumulator.
3. Perform the required bit manipulation using logical operations.
4. Store the result in the specified memory location.
5. Verify the output by comparing it with the expected result.

(a) Verification of Bit Masking Operations

Formula

Result = Input AND Mask

Input

- Input data = A5H
- Mask = 0FH

Code (8085 Assembly Language)

```
LXI H, 2000H
MOV A, M
INX H
ANA M
```

INX H

MOV M, A

HLT

Output

- Memory location 2000H = A5H
- Memory location 2001H = 0FH
- Memory location 2002H = 05H
- Result: A5H AND 0FH = 05H

EXECUTION:



(b) Testing of Bit Setting and Clearing Operations

(i) Bit Setting

Formula

Result = Input OR Mask

Input

- Input data = 40H
- Mask = 08H

Code (8085 Assembly Language)

```
LXI H, 2000H
MOV A, M
INX H
ORA M
INX H
MOV M, A
HLT
```

Output

- Memory location 2000H = 40H
- Memory location 2001H = 08H
- Memory location 2002H = 48H
- Result: 40H OR 08H = 48H

(ii) Bit Clearing

Formula

Result = Input AND (NOT Mask)

Input

- Input data = FFH
- Mask = FEH

Code (8085 Assembly Language)

```
LXI H, 2000H
MOV A, M
INX H
ANA M
INX H
MOV M, A
```

HLT

Output

- Memory location 2000H = FFH
- Memory location 2001H = FEH
- Memory location 2002H = FEH
- Result: FFH AND FEH = FEH

EXECUTION:



STUDENT REFLECTION

This experiment helped me understand bit manipulation techniques using 8085 assembly language. I learned to perform bit masking, set specific bits, clear selected bits, and verify the results using logical operations.

RESULT

Thus, the assembly language programs for bit masking, bit setting, and bit clearing were developed and verified successfully using the 8085 microprocessor instruction set.